SONAM RAJESH KORADE

Full Stack Web Developer | AI/ML Enthusiast

+91 8999695006 ✉ sonamkorade25@gmail.com LinkedIn | GitHub

PROFILE

Final-year B.E. Information Technology student at Walchand Institute of Technology, Solapur, with hands-on experience as a web developer and AI/ML developer building full-stack applications, LLM-integrated systems, and computationally intensive engineering projects. Skilled in React.js, Node.js, JavaScript, Python, and applied machine learning, with working knowledge of REST APIs, RAG pipelines, and browser-based ML inference. SIH 2025 Finalist with a proven track record of delivering production-grade projects at national-level competitions. Passionate about building scalable web systems and intelligent, AI-driven applications.

EDUCATION

Walchand Institute of Technology, Solapur — *Solapur, Maharashtra*
Bachelor of Technology in Information Technology; CGPA: 8.95 — 2023 – Pursuing (2027)

Rajarshi Shahu College, Latur — *Latur, Maharashtra*
12th Board (HSC); Percentage: 61% — 2022 – 2023

PROJECTS

MAITRI – AI Assistant for Astronaut Well-Being — *SIH Finalist 2025*

- Developed an AI-based system to detect stress and fatigue in astronauts using facial expressions and voice tone from real-time video and audio.
- Built a voice-activated assistant that provides mindfulness and grounding techniques, along with an admin dashboard to monitor mood and vital trends.
- Integrated REST APIs to collect and process sensor data, contributing to the project's selection as a Smart India Hackathon 2025 Finalist among thousands of teams.

Viraat – GPU-Accelerated Mathematical Optimization Solver — *Numerical Optimization · Sparse Linear Algebra*

- Built a sovereign mathematical optimization solver from scratch to solve Linear Programming (LP), Mixed-Integer Linear Programming (MILP), and Quadratic Programming (QP) problems at industrial scale.
- Designed a modular solver architecture incorporating presolve, simplex/interior-point methods, branch-and-bound, cutting planes, and heuristics for large, sparse, and highly constrained problems.
- Implemented sparse matrix computation techniques and GPU/multi-core acceleration to improve numerical stability and scalability.
- Applied the solver to real-world problem classes including refinery scheduling, production planning, logistics, and power-system dispatch.

Privacy-Preserving Visual Browser Agent — *WebGPU · WebAssembly · ONNX Runtime Web · Transformers.js*

- Built a browser-based AI agent that performs on-device visual perception to interpret screen context without exposing sensitive user data to a server.
- Designed a client-side privacy pipeline that detects and redacts sensitive elements — such as faces, passwords, and personal information — before any visual context leaves the device.
- Implemented lightweight local inference using WebGPU- and WebAssembly-based runtimes for real-time, on-device screen understanding.
- Integrated sanitized visual context with a server-side LLM/VLM to interpret user tasks and generate browser actions such as clicking, scrolling, and form interaction.

TECHNICAL SKILLS

Web Development: React.js, Node.js, Express.js, JavaScript, HTML, CSS, REST APIs, Responsive Web Development

AI/ML & Data: Machine Learning, Natural Language Processing (NLP), LLM APIs, RAG, Pandas, NumPy, Basics of Deep Learning

Programming & Development: Python, JavaScript, SQL

Design & Tools: UI/UX Design, Digital Illustration, Responsive Design, Git/GitHub, Figma

Professional Skills: Communication, Team Collaboration

CERTIFICATIONS

- Python Programming Certification | Infosys Springboard — Core Python, OOP, file handling, and data structures.
- Full Stack Web Development (MERN) | Udemy — HTML5, CSS3, JavaScript, React.js, Node.js, Express.js, and MongoDB.

ACHIEVEMENTS

- Smart India Hackathon (SIH) 2025 Finalist — Selected nationally for MAITRI, an AI-powered astronaut well-being monitoring assistant, competing against thousands of teams pan-India.
- Ideathon – Walchand Institute of Technology, Solapur — Presented an innovative technical solution and received recognition for problem-solving approach and prototype quality.